SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

QUESTIONS: 5 **Max Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

*I **A. Midhuna (192521302**) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

A. Midhuna

Signature of the student:

Annexure A

Scenario-Based Questions

1. **Dental X-Ray Enhancement and Cavity Region Extraction**

A dental clinic introduces a computer vision system to analyse panoramic dental X-ray images for early detection of cavities and bone loss. The X-rays suffer from low contrast, overexposed and underexposed areas, and quantum noise, making it difficult to separate decayed regions from healthy enamel and surrounding bone.

Question: Apply suitable spatial-domain enhancement techniques, such as log or power-law (gamma) transformations, histogram equalization, and median or Gaussian filtering, to improve the visibility of the suspected cavity regions. Subsequently, apply an appropriate global or adaptive thresholding or region-growing technique to extract the cavity regions from the enhanced image. Explain the sequence of processing steps and the role of each technique in obtaining the required region.

Answer:

Dental X-Ray Enhancement and Cavity Region Extraction

Aim

To enhance dental panoramic X-ray images using spatial-domain techniques and extract suspected cavity regions through thresholding.

Processing Steps

Read the Dental X-Ray Image

Load the dental X-ray image and convert it into grayscale if it is a colour image. Grayscale conversion simplifies image processing by representing each pixel with an intensity value.

Image Enhancement

a) Gamma Transformation

Gamma transformation improves the visibility of dark or bright regions in an image. By selecting a suitable gamma value, underexposed areas can be enhanced.

- Gamma < 1: Brightens the image.
- Gamma > 1: Darkens the image.

b) Histogram Equalization

Histogram equalization redistributes pixel intensity values to improve the overall contrast of the X-ray image. It helps distinguish dental structures and makes variations between healthy enamel and suspected cavities more visible.

c) Median Filtering

Median filtering removes impulse noise while preserving important edges. It helps reduce noise without excessively blurring the boundaries of suspected cavity regions.

d) Gaussian Filtering

Gaussian filtering smooths the image and reduces high-frequency noise. However, excessive smoothing should be avoided because it may blur small cavities.

Cavity Region Extraction Using Adaptive Thresholding

After enhancement and noise removal, adaptive thresholding is applied to separate regions based on local intensity variations.

Unlike global thresholding, adaptive thresholding calculates a threshold for each local neighbourhood. Therefore, it is more suitable for dental X-rays containing uneven illumination and overexposed or underexposed areas.

The resulting binary image highlights candidate regions that may contain cavities.

Morphological Processing

Morphological operations improve the extracted regions:

- **Opening:** Removes small noise particles.
- **Closing:** Fills small gaps and connects nearby pixels.
- **Connected-component analysis:** Identifies separate candidate regions for further examination.

Display the Results

Display the original X-ray, enhanced image, filtered image, and extracted binary regions. Candidate regions can also be outlined on the original X-ray for visual inspection.

Sequence of Processing

Input X-ray → Grayscale Conversion → Gamma Transformation → Histogram Equalization → Median/Gaussian Filtering → Adaptive Thresholding → Morphological Processing → Candidate Cavity Regions

Role of Each Technique

Technique	**Purpose**

Gamma transformation	Improves visibility in dark or bright regions
Histogram equalization	Enhances image contrast
Median filtering	Reduces noise while preserving edges
Gaussian filtering	Smooths the image and reduces high-frequency noise
Adaptive thresholding	Separates candidate regions under uneven illumination
Morphological processing	Removes noise and improves region boundaries
Connected-component analysis	Identifies individual candidate regions

Conclusion

Spatial-domain enhancement techniques improve the contrast and visibility of dental X-ray images. Median or Gaussian filtering reduces noise, while adaptive thresholding extracts candidate regions based on local intensity differences. Morphological operations refine the extracted regions, making them easier to examine.

2. **Printed Circuit Board (PCB) Fault Inspection**

An electronics manufacturer uses a high-resolution camera on its assembly line to inspect printed circuit boards for broken tracks, short circuits, missing solder joints, and hairline cracks. The captured images contain reflections from solder surfaces, sensor noise, and faint, thin fault boundaries that are easily lost in the background pattern of the board.

Question: Apply suitable image smoothing and sharpening techniques, such as averaging or Gaussian filters and Laplacian or unsharp masking, to improve the quality of the PCB images. Use appropriate point, line, or edge detection methods (for example Sobel, Prewitt, or Canny operators) to identify potential faults in the copper tracks and solder joints. Finally, apply edge linking and boundary detection techniques to obtain continuous fault boundaries. Describe the processing pipeline and explain how each selected technique contributes to fault identification.

Answer:

Printed Circuit Board (PCB) Fault Inspection

Aim

To enhance PCB images using smoothing and sharpening techniques, detect potential faults using edge detection methods, and identify continuous fault boundaries using edge linking and boundary detection.

Processing Steps

Step 1: Image Acquisition and Grayscale Conversion

Capture the PCB image using a high-resolution camera and convert it into grayscale. This simplifies processing by representing each pixel with an intensity value.

Step 2: Image Smoothing

Apply a Gaussian filter to reduce sensor noise and unwanted reflections.

- Averaging filter: Reduces noise by averaging neighbouring pixel values, but may blur thin fault boundaries.
- Gaussian filter: Smooths the image while reducing high-frequency noise. It is generally preferred for preparing PCB images for edge detection.

Step 3: Image Sharpening

Apply Laplacian filtering or unsharp masking to enhance the visibility of thin copper tracks, hairline cracks, and solder-joint boundaries.

- Laplacian filter: Highlights rapid intensity changes and fine details.
- Unsharp masking: Subtracts a blurred image from the original image and adds the resulting detail back to enhance edges.

Step 4: Edge Detection

Apply the Canny edge detector to identify potential faults in copper tracks and solder joints.

The Canny method involves noise reduction, gradient calculation, non-maximum suppression, and double thresholding with hysteresis.

Alternatively, Sobel or Prewitt operators can detect edges by calculating intensity gradients in horizontal and vertical directions.

These techniques help identify broken tracks, missing solder joints, short-circuit connections, and hairline cracks.

Step 5: Edge Linking

Use edge linking to connect nearby edge pixels that belong to the same fault boundary.

Edge linking can use gradient direction, pixel connectivity, or morphological operations to connect fragmented edges and improve boundary continuity.

Step 6: Boundary Detection

Apply contour detection to identify continuous boundaries around suspected fault regions. The detected contours can be overlaid on the original PCB image to highlight potential defects for inspection.

Processing Pipeline

Input PCB Image → Grayscale Conversion → Gaussian Filtering → Image Sharpening → Canny Edge Detection → Edge Linking → Boundary Detection → Fault Identification

Role of Each Technique	
Technique	Purpose
Grayscale conversion	Simplifies image processing
Gaussian filtering	Reduces sensor noise and unwanted variations
Laplacian filtering	Highlights fine details and intensity changes
Unsharp masking	Enhances thin tracks and crack boundaries
Canny operator	Detects clear and thin edges
Sobel/Prewitt operators	Detect horizontal and vertical intensity changes
Edge linking	Connects fragmented edge pixels
Boundary detection	Identifies continuous contours around suspected faults

Conclusion

Image smoothing reduces noise, while sharpening enhances faint fault boundaries. Edge detection identifies potential defects in copper tracks and solder joints. Edge linking and boundary detection connect fragmented edges and extract continuous fault boundaries, improving PCB inspection

3. **Underwater Image Enhancement and Marine Object Segmentation**

A marine research institute deploys an underwater drone to capture images of coral reefs, fish, and submerged debris. The images suffer from haze caused by light scattering, a strong blue-green colour cast, low brightness at greater depths, and periodic noise from the camera housing. The institute needs to extract meaningful regions such as coral patches and debris for further study.

Question: Apply suitable histogram processing and frequency-domain enhancement techniques to improve the quality of the underwater images. Select an appropriate low-pass, high-pass, or notch filtering technique based on the degradation characteristics, and justify how it reduces haze or periodic noise while retaining fine details. Subsequently, apply thresholding or region-based segmentation to separate coral

and debris regions from the water background. Explain the order of operations and how the selected techniques help extract the target regions.

Answer:

Underwater Image Enhancement and Marine Object Segmentation

Aim

To enhance underwater images using histogram processing and frequency-domain filtering, and to segment coral reefs and submerged debris from the water background.

Processing Steps

Step 1: Image Acquisition and Colour Correction

Load the underwater image and correct the blue-green colour cast by adjusting the colour channels. Improve brightness and colour balance to make coral and debris more distinguishable from the water.

Step 2: Histogram Processing

Apply histogram equalization or Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve image contrast.

- **Histogram equalization:** Redistributes intensity values to improve overall contrast.
- **CLAHE:** Enhances local contrast while limiting excessive amplification of noise.

CLAHE is particularly useful for revealing coral structures and submerged objects in low-contrast underwater images.

Step 3: Frequency-Domain Enhancement

Convert the image into the frequency domain using the Fast Fourier Transform (FFT). Select the filtering technique according to the type of degradation.

- **Low-pass filter:** Reduces high-frequency noise but may blur fine details.
- **High-pass filter:** Enhances fine details and object boundaries but may amplify noise.
- **Notch filter:** Removes specific periodic noise frequencies while preserving most other image information.

Since the image contains periodic noise from the camera housing, a **notch filter** is appropriate for suppressing identifiable periodic noise components in the frequency spectrum.

After filtering, apply the inverse FFT to reconstruct the enhanced image.

Step 4: Brightness and Contrast Enhancement

Apply gamma correction to improve the visibility of objects in darker underwater regions.

A suitable gamma value can brighten underexposed regions, making coral patches and submerged debris easier to identify.

Step 5: Image Segmentation

Apply thresholding or region-based segmentation to separate the target objects from the water background.

- **Thresholding:** Separates pixels based on intensity or colour values.
- **Region growing:** Starts from selected seed pixels and groups neighbouring pixels with similar colour or intensity.
- **Colour-based segmentation:** Uses colour information to distinguish coral and debris from surrounding water.

Region growing or colour-based segmentation is useful when target objects have similar brightness to the background but differ in colour or local appearance.

Step 6: Morphological Processing

Apply morphological operations to refine the segmented regions.

- **Opening:** Removes small unwanted noise regions.
- **Closing:** Fills small gaps and connects nearby regions.

Finally, use connected-component analysis or contour detection to identify individual coral patches and debris regions.

Processing Pipeline

Input Underwater Image → Colour Correction → CLAHE → FFT → Notch Filtering → Inverse FFT → Gamma Correction → Region-Based Segmentation → Morphological Processing → Object Extraction

Role of Each Technique

Technique	**Purpose**
Colour correction	Reduces the blue-green colour cast
Histogram equalization / CLAHE	Improves contrast and reveals details
FFT	Converts the image into the frequency domain
Notch filtering	Removes specific periodic noise frequencies
Low-pass filtering	Reduces high-frequency noise
High-pass filtering	Enhances fine details and boundaries
Gamma correction	Improves visibility in dark regions

Thresholding	Separates objects based on intensity or colour
Region growing	Groups neighbouring pixels with similar properties
Morphological processing	Removes small regions and fills gaps
Contour detection	Identifies object boundaries

Conclusion

Histogram processing improves contrast, while frequency-domain filtering reduces periodic noise and enhances useful image details. Colour correction and gamma transformation improve the visibility of underwater objects. Thresholding or region-based segmentation separates coral patches and submerged debris from the water background, while morphological operations refine the extracted regions.

4.**Railway Track Fault Detection Using Inspection Cameras**

A railway inspection vehicle captures images of tracks during early morning, bright daylight, and in tunnels. The images contain strong shadows from overhead structures, uneven illumination, gravel and ballast texture, the two rail lines, and occasional rail cracks or missing fasteners. The vision system must extract the rail boundaries and identify visible cracks before further analysis.

Question: Apply suitable grey-level transformations and spatial filtering techniques to improve the track images under varying illumination conditions. Use thresholding and edge detection to extract rail lines and crack candidates. Apply line detection (such as the Hough transform), edge linking, and boundary detection wherever appropriate to obtain continuous rail edges and crack outlines. Present the sequence of processing operations and explain the purpose of each technique in extracting the required image regions.

Answer:

Railway Track Fault Detection Using Inspection Cameras

Aim

To enhance railway track images under varying illumination conditions, detect rail lines and crack candidates using thresholding and edge detection, and extract continuous rail boundaries using the Hough transform and edge linking techniques.

Processing Steps

Step 1: Image Acquisition and Grayscale Conversion

Capture railway track images using an inspection camera and convert them into grayscale. This simplifies processing by representing each pixel using its intensity value.

Step 2: Grey-Level Transformations

Apply suitable grey-level transformations to improve visibility under different lighting conditions.

- Gamma transformation: Corrects dark images captured in tunnels or early morning. A gamma value less than 1 brightens dark regions.
- Histogram equalization: Improves overall contrast by redistributing pixel intensity values.
- CLAHE: Enhances local contrast and helps reveal rail edges and cracks under uneven illumination without excessively amplifying the entire image.

Step 3: Spatial Filtering

Apply spatial filters to reduce noise and enhance important details.

- Gaussian filtering: Reduces noise caused by gravel texture and camera sensors while smoothing the image.
- Median filtering: Removes impulse noise while preserving important edges.
- Laplacian filtering or unsharp masking: Sharpens rail boundaries and improves the visibility of fine cracks.

Gaussian filtering is generally applied before edge detection to reduce unwanted edges caused by noise.

Step 4: Thresholding

Apply global or adaptive thresholding to separate the rails and other relevant structures from the background.

- Global thresholding: Uses a single threshold for the entire image and is suitable when illumination is relatively uniform.
- Adaptive thresholding: Calculates local thresholds and is more suitable for images containing shadows and uneven illumination.

Thresholding helps identify candidate rail regions, but gravel and ballast may also be selected. Therefore, further processing is required.

Step 5: Edge Detection

Apply the Canny edge detector to identify rail boundaries and crack candidates.

The Canny operator detects intensity changes while reducing false edges through non-maximum suppression and double-threshold hysteresis.

Sobel or Prewitt operators can also be used to detect horizontal and vertical intensity gradients.

Step 6: Rail Line Detection Using the Hough Transform

Apply the Hough line transform to the detected edges to identify straight or nearly straight rail boundaries.

The Hough transform identifies lines by analysing the possible combinations of line parameters supported by edge pixels.

Since railway rails extend along the track, the detected lines can be selected according to their orientation, position, and expected spacing.

This helps distinguish the main rail boundaries from irregular gravel edges.

Step 7: Edge Linking and Boundary Detection

Apply edge linking to connect fragmented rail edges and improve their continuity.

Use contour detection or boundary tracing to extract the outlines of the rails and candidate crack regions.

Morphological closing can help connect small gaps in rail boundaries, while opening can remove small isolated noise regions.

Step 8: Crack Candidate Identification

Examine the detected edge patterns for short, irregular discontinuities or thin line structures that differ from the normal rail surface.

Use local contrast, shape, orientation, and position to distinguish potential cracks from normal rail edges and surface texture.

The resulting crack candidates can be highlighted on the original image for further inspection.

Processing Pipeline

Input Image → Grayscale Conversion → Gamma Correction → CLAHE → Gaussian/Median Filtering → Adaptive Thresholding → Canny Edge Detection → Hough Line Transform → Edge Linking → Boundary Detection → Crack Candidate Identification

Role of Each Technique

Technique	Purpose
Grayscale conversion	Simplifies image processing
Gamma transformation	Improves visibility in dark images
Histogram equalization	Improves overall contrast
CLAHE	Enhances local contrast under uneven illumination
Gaussian filtering	Reduces noise and unwanted texture
Median filtering	Removes impulse noise while preserving edges

Laplacian/unsharp masking	Enhances fine details and crack boundaries
Adaptive thresholding	Separates candidate regions under varying illumination
Canny edge detection	Identifies rail edges and crack candidates

Conclusion

Grey-level transformations improve image visibility under different lighting conditions, while spatial filtering reduces noise and enhances important details. Adaptive thresholding and Canny edge detection identify rail boundaries and potential cracks. The Hough transform detects the main rail lines, and edge linking and boundary detection improve the continuity of the extracted outlines. Note: These techniques identify potential cracks rather than confirming defects. Further analysis using crack geometry, reference images, or a trained detection model may be required to distinguish actual rail faults from shadows, surface marks, and ballast texture.

5. **Fruit Quality Grading Through Bruise Region Extraction**

A fruit processing plant uses a conveyor-mounted camera to grade apples by detecting bruises, rot spots, and surface blemishes. The images contain glossy highlights from the fruit skin, shadows between adjacent fruits, the conveyor belt background, and dark bruise regions that have low contrast and irregular shapes against the natural colour variation of the fruit.

Question: Apply suitable grey-level or contrast enhancement techniques to improve the visibility of bruised and rotten regions. Use an appropriate smoothing filter to suppress noise and specular highlights while preserving the fruit outline. Subsequently, apply thresholding or region-based segmentation to separate the fruit from the background and to isolate suspected bruise patches. Use edge detection and boundary detection to identify the outlines of the extracted regions. Explain the processing sequence and the contribution of each technique to bruise-region extraction.

Answer:

Fruit Quality Grading Through Bruise Region Extraction

Aim

To enhance fruit images, reduce noise and glossy highlights, separate fruits from the background, and extract suspected bruise and rot regions using image segmentation and boundary detection techniques.

Processing Steps

Step 1: Image Acquisition and Colour Conversion

Capture the fruit image using a conveyor-mounted camera. Convert the image from RGB to HSV or grayscale, depending on the segmentation method.

HSV colour space is useful because it separates colour information from brightness, helping distinguish bruised regions from healthy fruit skin.

Step 2: Grey-Level and Contrast Enhancement

Apply suitable enhancement techniques to improve the visibility of dark bruise and rot regions.

- Histogram equalization: Improves the overall contrast of the image.
- CLAHE: Enhances local contrast and makes low-contrast bruise regions more visible.
- Gamma transformation: Adjusts image brightness. A gamma value less than 1 brightens dark regions.

CLAHE is particularly useful for highlighting subtle blemishes without excessively changing the entire image.

Step 3: Image Smoothing

Apply a suitable smoothing filter to reduce noise while preserving fruit boundaries.

- Gaussian filter: Reduces sensor noise and small intensity variations.
- Median filter: Removes impulse noise while preserving important edges.

A median filter can help suppress small bright or dark artifacts, but strong specular highlights may require additional highlight detection and correction. Smoothing alone cannot reliably remove all reflections.

Step 4: Fruit Segmentation

Separate the fruit from the conveyor belt using colour-based segmentation or thresholding.

- Colour thresholding: Uses HSV colour ranges to identify fruit pixels.
- Adaptive thresholding: Separates regions based on local intensity variations.
- Morphological opening: Removes small unwanted background regions.
- Morphological closing: Fills small gaps in the segmented fruit.

Connected-component analysis can then identify individual fruits, even when multiple fruits appear in the same image.

Step 5: Bruise and Rot Region Extraction

After separating the fruit from the background, analyse only the pixels inside the fruit region.

Apply colour-based or region-based segmentation to identify suspected bruises and rot spots.

- Colour-based segmentation: Identifies regions with colour characteristics different from healthy fruit skin.

- Adaptive thresholding: Detects local dark regions when brightness varies across the fruit surface.
- Region growing: Groups neighbouring pixels with similar colour and intensity, helping extract irregularly shaped blemishes.

Using both colour and local contrast can improve segmentation because some bruises may not be significantly darker than the surrounding skin.

Step 6: Edge Detection

Apply the Canny edge detector to identify the boundaries of suspected bruise and rot regions.

The Sobel operator can also be used to calculate horizontal and vertical intensity gradients.

These methods highlight intensity transitions between healthy fruit skin and suspected damaged regions.

Step 7: Boundary Detection

Apply contour detection to extract the outlines of the segmented bruise regions.

Morphological closing can connect small gaps, while opening can remove isolated noise components.

Finally, overlay the detected boundaries on the original image to visualise the suspected damaged areas.

Processing Pipeline

Input Image → HSV/Grayscale Conversion → CLAHE → Median/Gaussian Filtering → Fruit Segmentation → Bruise Region Extraction → Canny Edge Detection → Morphological Processing → Boundary Detection

Role of Each Technique

Technique	Purpose
HSV colour conversion	Separates colour information from brightness
Histogram equalization	Improves overall contrast
CLAHE	Enhances local contrast in bruise regions
Gamma transformation	Adjusts image brightness
Gaussian filtering	Reduces noise and small intensity variations
Median filtering	Removes impulse noise while preserving edges
Colour thresholding	Separates fruit from the background
Adaptive thresholding	Detects dark regions under uneven illumination
Region growing	Extracts connected regions with similar properties

Conclusion

Contrast enhancement improves the visibility of bruises and rot spots, while smoothing filters reduce noise and preserve important fruit boundaries. Colour-based segmentation separates the fruit from the conveyor belt, and adaptive thresholding or region growing extracts suspected damaged areas. Edge detection and contour analysis identify the outlines of these regions for fruit quality grading.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand